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Perspective

Ten Problems in Polymer Reactivity Prediction

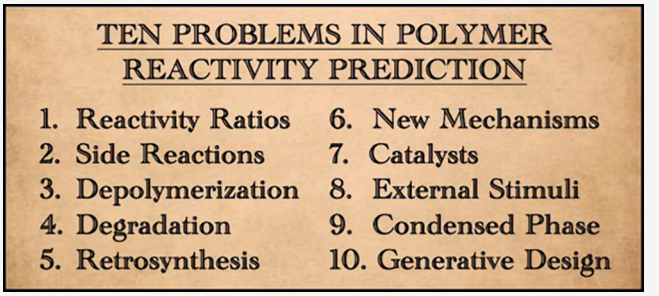
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ABSTRACT: The phrase "polymer theory" is typically associated with classical predictions of structural, thermodynamic, or mechanical properties in which chemical structure is treated nonspecifically. This is because historically, modeling chemically specific polymer phenomena or predicting their properties required prohibitively costly computational methods. However, recent methodological developments have made previously inaccessible polymer reactivity predictions now potentially addressable with a combination of data-driven and physics-based computational approaches. This perspective highlights ten such problems, showcases related previous successes, and speculates on what the near future holds as working solutions to polymer reactivity prediction become available.



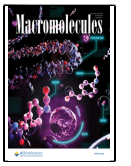
INTRODUCTION

Though the current state of commercial polymers invites cynicism,[1] biology reminds us that precisely controlled and sustainably synthesized polymers can be created with negligible environmental impact.[2−4] Critical to accessing this space of macromolecular design is the elucidation of robust synthetic pathways for polymers that proceed with chemical specificity and high structural fidelity.[5] Provided the difficulty of accessing finely detailed reaction mechanism information experimentally, the field of polymer synthesis stands to benefit from advances in computational molecular science. However, the theoretical and computational apparatus that has historically pervaded polymer science has been strongly biased by a community that treats molecular structure as a fitting parameter for less-detailed models at mesoscopic length scales.[6−9] This is in opposition to the goals of many areas of polymer research where understanding and controlling the electronic structure or reactivity of specific chemistries are central.

In the past two decades circumstances have changed with respect to the goals of polymer science and the accessible quantum mechanical methods for molecular prediction. In contrast to the 20th century polymer science goals of optimizing thermodynamics, mechanical properties, and rheology, the current zeitgeist of polymer science has shifted toward concerns of sustainability, degradation, nonpetroleum sourcing pathways, and environmental impacts.[10] Well-timed with this change in focus for the polymer community has been the dramatic evolution of quantum chemistry (QC) and machine learning (ML) methods for describing organic reactivity that are required to credibly address these design challenges. First-principles, quantum mechanical reaction pathway predictions for arbitrary organic molecular structures have become common place. Methods are appearing that integrate reaction prediction with microkinetic models to explore longer reaction sequences and reaction networks. In parallel, a broad range of ML model architectures and methods have become available to complement purely physics-based approaches. These advances create an opportunity to revisit the problem of polymer reaction prediction with the goal of delivering broadly impactful methods capable of transforming polymer science.

This is not to say that computational reactivity prediction in polymer science has not been productively employed to-date.[11−13] While the number of papers utilizing a QC treatment of polymer reactivity is far exceeded by the number of papers utilizing classical atomistic or coarse-grained simulations, this body of work has provided essential insights into the understanding of polymerization mechanisms. Several groups have characterized reversible addition−fragmentation chain-transfer (RAFT) polymerization,[14−27] atom transfer radical polymerization (ATRP),[27−32] and nitroxide-mediated polymerization (NMP),[33−39] while establishing computational best practices.[40,41] A number of groups have employed QC to investigate the metal-catalyzed polymerization of olefins.[42−67] Broadbelt, Moscatelli, Rappe, and others have contributed extensively in the application of density functional theory (DFT) for radical-mediated reactivity in polymers,[11,68−76] reactivity-ratio predictions,[70,77−83] the estimation of ceiling

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temperatures,[84] and the kinetics of polymerization[83,85−87] and decomposition.[88] Relatedly, a number of efforts have also employed QC to study Ring-Opening Polymerization (ROP) processes.[83,89−119] It is also critical to note the significant body of work from computational organic chemists on small molecule transformations[120] from which many advances in polymer synthesis are derived. For instance, ring-opening metathesis polymerization studies have benefited from a vast literature on small molecule studies of cross-coupling, ring-opening, and ring-closing reactions that are often supported by computational efforts. To this end, there are likely many important reactivity problems in polymer chemistry for which at least partial understandings and/or solutions already exist in the small molecule chemistry literature, even if these have yet to be fully exploited in polymer science. Taken together, these efforts showcase that QC is sufficiently mature to contribute quantitatively to the prediction and design of polymer synthesis and reactivity.

Notwithstanding the existing success of QC in radical-mediated polymerizations, applications of QC to the broader scope of polymer reactivity and synthesis in the community are more limited. Little work has applied QC to the understanding and design of step-growth polymerizations,[121−128] with which more diverse chemical spaces are accessible. Similarly, there are few works on the QC modeling of metal-catalyzed cross-coupling polymerizations,[129] catalyst transfer polymerizations, or ionic polymerizations. Even within the context of traditional vinyl chemistry, less than 30 distinct chemistries have been modeled with QC to-date,[12,13] which represents a minute fraction of the accessible chemical space of polymers (let alone vinyls!). Perhaps most importantly, the dispersity and molecular weights of most polymerization reactions are limited by side reactions (e.g., chain transfer to solvent/monomer/chain transfer agents,[130] propagation of midchain radicals, β-scission,[131] backbiting,[131] cyclization). Although some rudimentary side-reaction mechanisms are available,[132] they are nonspecific and mainly used for rationalization rather than for prediction. Some works have begun to use QC to explore how secondary reactions influence polymerization efficiency,[61,133−139] but they are limited in number, and obtaining such information from experimental characterizations is challenging. Taken together, it appears that there is enormous potential progress to be made on these topics using well-established methodologies in QC predictions, which could have a direct impact on the experimental control of polymer reactivity.

It is crucially important to highlight just how impactful a quantitative understanding of molecular reactivity is to the prediction of larger scale polymer features and properties that occupy the interests of the broader polymer science community. For example, Flory's principle of equal reactivity has led to critical derivations such as the Poisson distribution for living polymerizations and the Flory−Schulz distribution for step growth polymerizations that are ubiquitous in polymer science and commonly inform polymer design. Even when analytical models such as these are not applicable, explicit modeling of polymer structure using molecular reactivities and numerical methods[13,140,141] have and will continue to provide crucial insights into network formation, copolymerization, and polymer deconstruction mechanisms. These computational methods that use molecular reactivities as inputs work synergistically with QC studies to provide quantitative prediction frameworks across the field. These tools will continue to inform the field's understanding of polymer reactivity and degradation and will benefit from the advancement of molecular reactivity prediction tools. We believe that the integration of QC reactivity prediction tools with statistical and numerical models of polymer structure will transform modern polymer science.

With these points in mind, the main purpose of this perspective is to foreground some opportunities in predictive polymer synthesis and reactivity that are afforded by modern QC and ML methods. This is not to imply that the focus of this perspective is an "either-or" situation when contrasted to traditional theoretical and computational polymer physics; it is simply pointing out the strong bias of the landscape of polymer modeling away from reactivity and synthesis predictions. For example, in two recent (and otherwise excellent!) reviews by Gartner[9] and Schmid,[142] the modeling of polymer reactivity and synthesis does not feature in the broader roadmap for the field, and is mentioned only in passing as a handful (out of >500) citations in both cases. Moreover, polymer physics stands to benefit immensely from a quantitative understanding of reactivity, as the fine control of molecular weight, dispersity, and branching can enable experimental manifestations of idealized systems necessary to test traditional polymer theory's own grand challenge problems, many of which have been recently enumerated by Jayaraman and Klok.[143] While the accurate control over molecular weight distribution,[144] composition, sequence, tacticity, topology, and termination chemistry have historically been left to Edisonian optimization, these properties are now within the remit of rational computational prediction. In what follows we briefly summarize the state-of-the-art in computational advances relevant to polymer reactivity prediction, and then proceed to discuss the future potential of this field through the identification of ten challenge problems for computational polymer synthesis and reactivity.

PROGRESS IN RELEVANT COMPUTATIONAL METHODS

The following section is brief and necessarily not comprehensive. We have grouped methodological developments into two topics, QC estimates of reaction energetics and reaction network enumeration, and selected only those recent examples that are most relevant to potentially addressing the ten problems we have chosen to highlight. Notably, almost none of these computational advances have occurred with polymers in mind.

Several parallel developments have occurred that dramatically reduce the cost, improve the accuracy, and increase the throughput of QC calculations needed to understand reactivity. These developments indirectly push the frontier of what is possible in polymer reactivity predictions. The development of modern semiempirical (e.g., GFN2,[145] B97−3c[146]) and general ML potentials (e.g., ANI[147]) have dramatically reduced the cost of generating approximate transition states and searching for distinct reaction conformers.[148] Hybrid approaches that use low-level methods to converge approximate transition states that are then refined at a higher level have increased the size of systems for which usefully accurate transition states can be localized.[149−155] Recent efforts in generative AI have also made progress in moving the needle in transition state estimation.[156,157] Although these examples are still limited in scope and accuracy, AI approaches will only improve with time and it is reasonable to plan for them becoming competitive with more expensive physics-based approaches. Even historically pathological reaction scenarios for electronic structure are becoming

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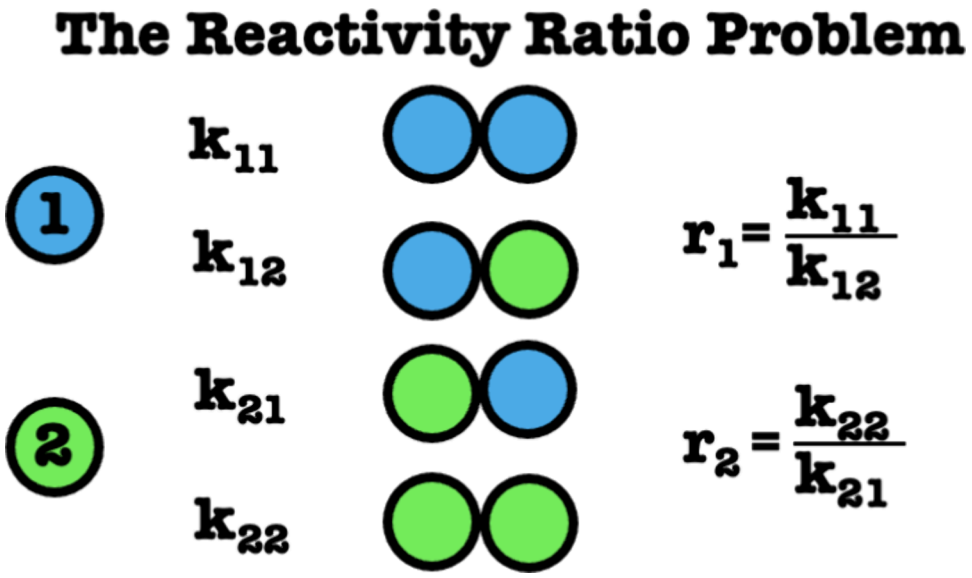


Figure 1. The reactivity ratio problem.

tractable for nonspecialists. Representative of these developments, recent work by Wardzala has illustrated the automation of multireference reactivity calculations, removing a long-standing computational barrier to describing pathological radical reactivity.[158] With this exceptional progress, there are of course gaps in the predictive capabilities of these methods that are important to acknowledge. Even for ground-state thermal chemistry, the accuracies of DFT functionals are insufficient for quantitative kinetic descriptions of deep reaction networks and reactions involving many transition metals. The high cost of excited-state QC has also made exploratory photoreactivity studies specialized exercises that do not inform routine reaction rationalization and optimization. Nevertheless, here too promising advances are underway,[159−162] but there is still a long way to go to meet the demand of a robust literature on photoinduced polymerization control from the polymer community.[163−165]

As the prediction of single-step reaction outcomes becomes more routine, there are exciting opportunities to combine this capability with network exploration algorithms to characterize multistep chemical reaction networks (CRN).[166−175] A single-step reaction exploration methodology is a prerequisite for deeper network exploration. In principle, recursive application of a single-step exploration methodology to the most favorable products of each reaction step constitutes a complete algorithm for exploring deeper networks. However, this naïve scenario soon runs up against the exponential growth of potential products that will stall exploration for even the most inexpensive reaction exploration methods. Thus, research is ongoing to develop algorithms that prioritize reactions for exploration to avoid this bottleneck. The field of chemical reaction network exploration is at an intermediate phase of maturity,[166,176] with ample demonstrations of deep networks, but also relatively few examples of quantitative predictions based solely on QC calculations. For polymer reactivity prediction, this means that such methods could plausibly be applied to predict multistep reaction deconstruction, upcycling, and polymerization scenarios, but only qualitatively describe deeper degradation sequences. However, these assessments are hypothetical because there has been remarkably little experimentation by the polymer community with the reaction and chemical reaction network exploration tools being developed by the reaction prediction community.

TEN PROBLEMS IN POLYMER REACTIVITY PREDICTION

Each of the following subsections are dedicated to *defining a problem in polymer reactivity*, sketching an overview of potential strategies, and providing a list of **examples that would constitute success**. Our selection intentionally eschews problems that align closely with traditional polymer property predictions (e.g., morphology, mechanics, chain conformations), and attempts to emphasize those most relevant to polymer reactivity and synthesis.

1. The Reactivity Ratio Problem. Statement: *Provided a set of regiochemical, stereochemical, or copolymerization outcomes, estimate the associated reactivity ratios needed to quantitatively describe the statistical yields.*

Synopsis: Reactivity ratios underlie the fundamentally statistical nature of all modern polymer synthesis (Figure 1). Whether it is sequence correlations in statistical copolymerizations,[177−179] or tacticity in homopolymerizations, controlling the reactivity ratios of different regiochemical or stereochemical outcomes is essential. While a modest amount of experimental data exists for traditional vinyl reactivity ratios,[180−184] it is a near ubiquitous experience for experimental collaborators to inquire if a computational scientist is capable of predicting reactivity ratios, as their experimental measurement is laborious and time-consuming. While predicting reaction rates necessarily requires the prediction of molecular transition states, the fact that rate prefactors will approximately cancel for reactivity ratios provides hope for a QC driven approach. Proof-of-principle for this problem has been achieved in a small space of vinyls,[70,77−83] and ML models have begun to emerge for the task of predicting radical propagation rates from experimental data.[185,186] However, the need for a chemically transferable computational tool capable of these estimation tasks across the broader chemical space of polymers is clear. Moreover, while less common than traditional radical copolymerization, anionic[187] and cationic[188] copolymerizations are also essential, with little pre-existing work on predicting their associated reactivity ratios with QC. The successes of using QC to predict reactivity ratios

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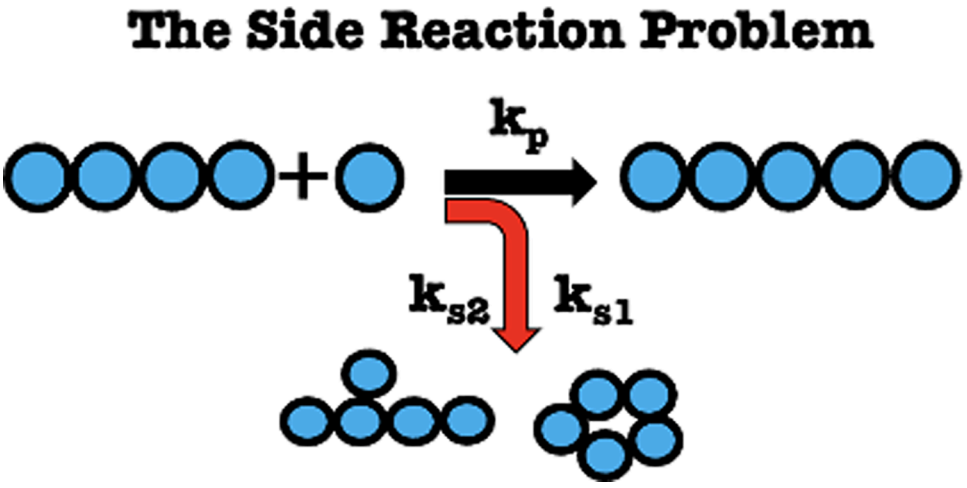


Figure 2. The side reaction problem.

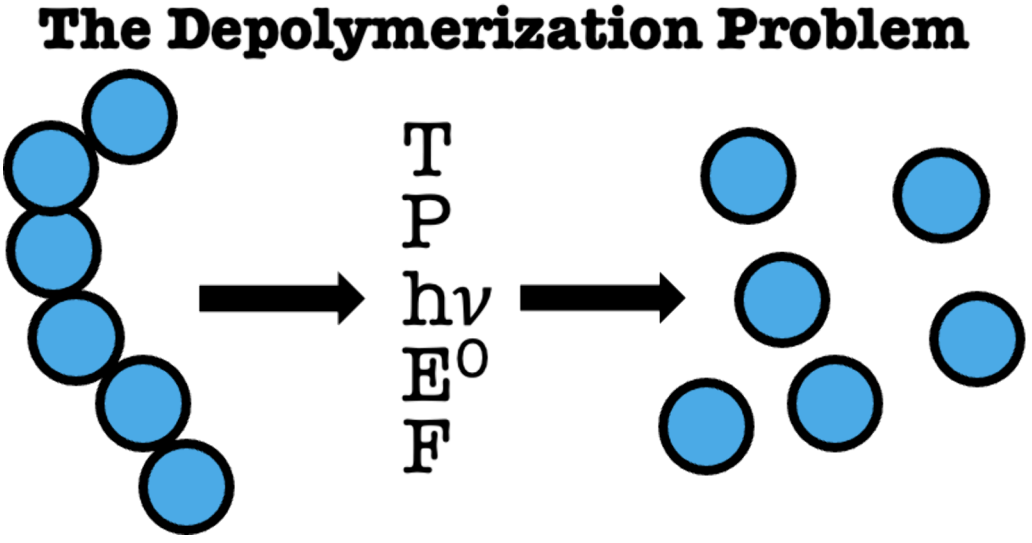


Figure 3. The depolymerization problem.

in narrow chemical spaces makes us optimistic that near-term pathways to generally accurate predictions might involve straightforward scale-up of QC-based data generation combined with emerging graph-based ML regressors. In the long term, the precise chemical control of reactivity ratios has the potential to push synthetic polymer chemistry into the realm of sequence control (at least, stochastically), which has remained a holy grail of polymer chemistry.[189]

Examples of Success:

- Create a computational model capable of taking as input multiple monomer chemical structures, and outputting all associated reactivity ratios in the solution-phase.
- Create a computational model that can compute reactivity ratios between different stereochemical outcomes, and is robust across chemical space.
- Clarify the prevalence of the penultimate effect across chemical space.

2. The Side Reaction Problem. Statement: *Provided a set of polymerization chemistries (monomer, solvent, catalyst, initiator, known impurities), elucidate all side reactions that occur at standard conditions.*

Synopsis: Knowing the scope of potential side reactions prior to entering the laboratory is a grand challenge for polymer chemistry (Figure 2). Side reactions are exceedingly challenging to access experimentally as they often only appear indirectly as modifications to primary reaction rate measurements accessible via standard experimental methods. Chain transfer, backbiting, beta-scission, and cyclization reactions represent a small subset of known inefficiency mechanisms for polymerization reactions, with most such knowledge derived from olefins and cyclic heterocycles. More broadly, many polymerization reactions are highly sensitive to impurities (e.g., anionic polymerization and oxygen/water), the side reactions of which are difficult to predict *a priori*. Even in the absence of impurities, a simple radical polymerization such as polyvinyl chloride exhibits at least 10 distinct radical types that contribute to the polymerization process.[190,191] The elucidation of all contributing pathways presents one of the more daunting prediction challenges in polymer chemistry, and one well-suited for advances in computational reaction networks. Luckily, the key for side reaction predictions is their relative rates, and how these relative rates compete with the rate of polymerization. Note that the solution of **The Reactivity Ratio Problem** represents a critical precursor toward addressing **The Side Reaction Problem**. To-

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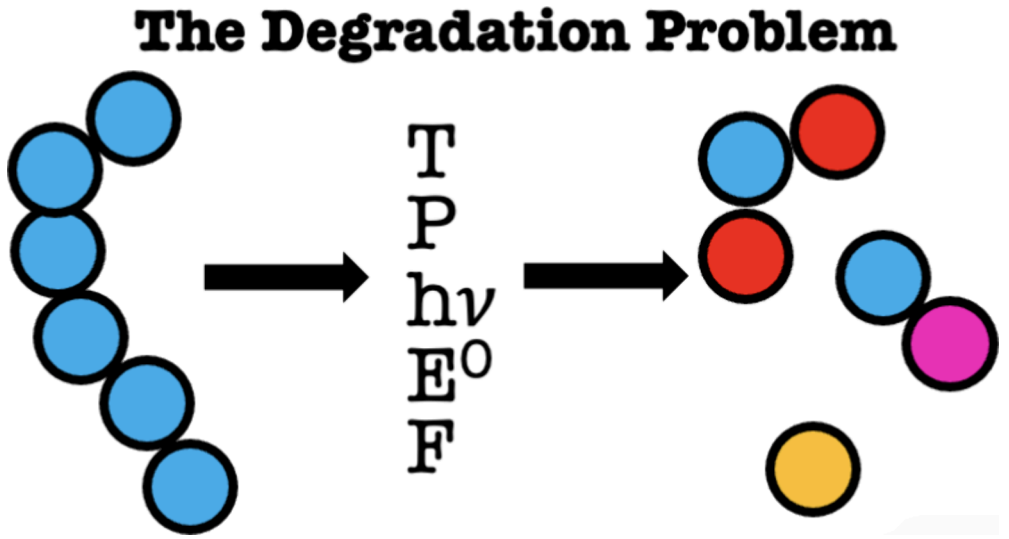


Figure 4. The degradation problem.

date, controlling side reaction chemistries experimentally has led to the state-of-the-art advances in controlled radical polymerizations, which foreshadows the potential impact of solutions to this problem. Given that many side-reactions are single step, this problem is potentially highly susceptible to reaction prediction methods that have become common place in other contexts (see Introduction). Thus, practical solutions to this problem may already exist with some minor adaptations but have yet to be benchmarked by the polymer community. More broadly, such considerations of controlling side reactions are particularly important for step-growth polymerizations and an efficient computational approach for their screening could prove game-changing.

Examples of Success:

- Discover new chemistries for controlled radical polymerizations that exceed existing standards.
- Quantitatively estimate the likelihood of homocouplings and beta couplings in e.g. direct arylation polymerization.
- Perform a retrospective computational analysis of controlled radical polymerization[192] chemistries using QC.
- Predict the ultimate products and side products of an electropolymerization reaction.
- Computationally enumerate the network of possible side reactions in a step growth polymerization.
- Identify all possible chain transfer reactions from a set of monomer, solvent, and catalyst.

3. The Depolymerization Problem. Statement: *Provided a monomer and polymer, identify the optimal reaction conditions for recovering the monomer.*

Synopsis: Controlled depolymerization is a direct pathway for polymer reuse and material circularity (Figure 3). However, at least three methodology gaps need to be addressed to make this possible. First, the thermodynamics of depolymerization to monomer and oligomers for specific polymer chemistries need to be made predictable. For practical purposes, this is summarized by being able to predict the ceiling temperature of a polymer as a function of state (e.g., in solution, melt, or specific morphology). The enthalpic contributions can likely already be predicted with sufficient accuracy given available methods like additivity theory or electronic structure calculations of model species;[84,193] but no analogous entropy description is sufficiently routine or accurate to account for condensed phase effects. Second, the kinetics of depolymerization are relevant to the design and optimization of any practical depolymerization process. The kinetics are at a minimum a function of the activation energies associated with the depolymerization reaction. In an uncatalyzed scenario, this may involve a small number of steps, although these may be quantitatively impacted by the details of polymer packing. In a catalyzed scenario, the relevant rate-limiting step may be as varied as the catalysts involved. In recent years, the accuracy, throughput, and reliability of transition state calculations have all increased which makes relative kinetic predictions already possible for depolymerizations with relatively well-defined steps. A more comprehensive answer to this subproblem would deliver quantitative (rather than relative) kinetics and would be able to address depolymerizations without established pathways. Third, depolymerization is always in competition with degradation chemistry, and thus there is strong overlap with **The Degradation Problem**. This is trivially seen by considering that every polymer thermodynamically favors depolymerization at its ceiling temperature, yet not all polymers can be thermally depolymerized without competing pathways to species besides monomer. Predicting depolymerization yield necessarily requires a consideration of one or more competing off-target pathways. The ability to codesign polymer chemistry and depolymerization profile would eliminate a costly Edisonian step from the materials development cycle. Control may also afford new polymer functionalities. For example, controlling the polymerization-depolymerization equilibrium is adjacent to designing *repolymerization* pathways. Dynamically adaptable bonds in polymer networks are a popular present research direction, but the ability to design a system that could depolymerize, and then repolymerize along a different motif optimized toward a stimuli response would be transformative.

Examples of Success:

- Develop an efficient protocol for estimating the entropy change in a polymerization that incorporates a broad range of conformations, enabling more accurate predictions of ceiling temperatures.
- Screen organic chemistries to find "sweet spots" in the thermodynamics and kinetics of reactivity that would

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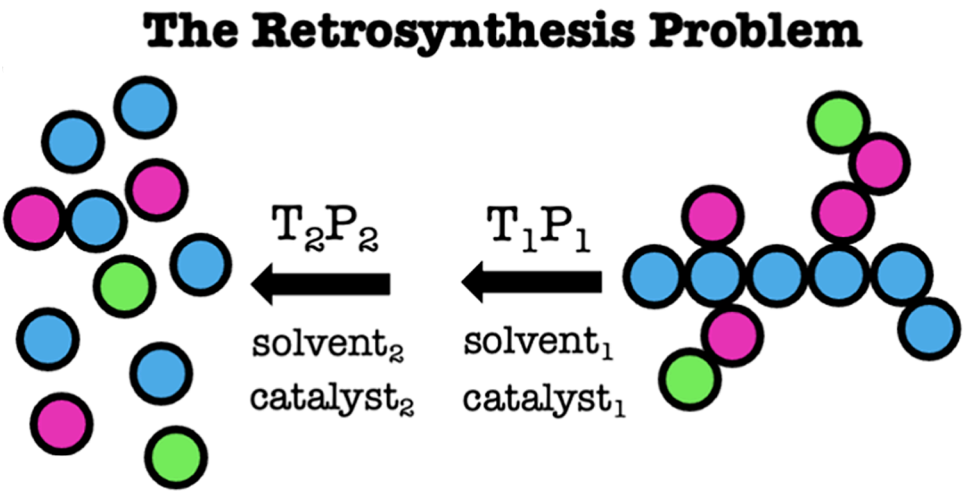


Figure 5. The retrosynthesis problem.

enable controlled polymerization, depolymerization, and repolymerization.

- Use computation to predict optimized depolymerization conditions that increase the yield of monomer.
- Leverage QC-driven reaction networks to optimize electrodepolymerization processes.

4. The Degradation Problem. Statement: *Provided a polymer and stressor, predict its terminal degradants and degradation pathways.*

Synopsis: While making the materials of tomorrow we also need to avoid making the forever chemicals of tomorrow. A large fraction of polymer chemists are currently involved in developing methods to degrade or upcycle legacy plastics. To avoid a repetition, we need methods for predicting or accelerating the characterization of how new plastics and their precursors degrade (Figure 4). The state of the art for determining the degradation time scale and identity of degradants is still make and break experimental testing. This is both extremely costly and minimally informative. Moreover, it only provides feedback at the end of a long scale-up process. This lofty prediction goal can be broken into a few subproblems. First, there is a potentially rich scope for hybrid experimental and computational methods of characterizing degradation. Such methods could rely on small sample size experiments and modeling to screen for the presence of particular degradants or rates. Although these would represent accelerated characterizations, rather than computational predictions, they could have a similar practical effect. Second, methods for predicting degradation pathways must be able to address the broad scope of potential stressors that mediate degradation. Depending on application or the environment, the relevant stressor may be radiative, oxidative, thermal, mechanical, or other stressors, and the predictability will vary. For example, computational methods for predicting reaction pathways are more mature for thermal chemistry than for radiative chemistry, which often involves multiple spin states and are more costly to access due to the need for excited state potential energy surfaces. Third, the interaction between stressors can open up new degradation pathways that are kinetically or thermodynamically disfavored when only considering single-stressor chemistry.

The potential range of degradation pathways may appear overwhelming, but polymer chemistry benefits from its generally contracted chemical scope and the fact that degradation chemistry is relatively local. This creates the opportunity to potentially database and define templates for common degradation susceptibilities that might accelerate the characterization of new variants. The diversity of possible chemistries is also the reason why most regulation is reactionary rather than proactive—you cannot regulate what you cannot define or characterize. For this reason, the ability to predict plausible degradants may also have profound implications for how plastics and their precursors are regulated.

Examples of Success:

- Predict the major pyrolysis products of commodity homopolymers. Thermal stress and homopolymers are probably the simplest starting combination.
- Predict the major degradants of a commodity polymer as a function of stressor.
- Predict multistep reaction pathways to terminal degradants.
- Computationally optimize metabolic pathways to decompose commodity polymers in bioreactors.

5. The Polymer Retrosynthesis Problem. Statement: *Provided a targeted polymer structure, predict a monomer and polymerization scheme that is optimal with respect to applicable restraints.*

Synopsis: The retrosynthesis problem has begun to be broached across the field of organic chemistry,[194] but a robust solution remains elusive (Figure 5). Provided a polymer with a targeted chemistry and macromolecular architecture, one must be able to invert the statistical polymerization process to associated reactivity ratios, side reactions, and reaction rates, all of which will be solvent, monomer, catalyst, and thermodynamically dependent. This challenge is rendered difficult by the many competing polymerization strategies for chain growth, including (but not limited to) cationic, anionic, conventional radical, ATRP, RAFT, and NMP. For all of these polymerization strategies, a collection of heuristic guidelines have been accumulated over the years: monomers with electron withdrawing/donating groups (EWGs/EDGs) are more suitable for anionic/cationic polymerization, respectively. Cationic polymerizations require polar, but non-nucleophilic solvents, and usually only work at low temperatures. Anionic polymerizations are very impurity sensitive (e.g., water, oxygen). ATRP is tolerant to a wide range of chemistries and exhibits great MW control, but it is often difficult to do ATRP in aqueous media and

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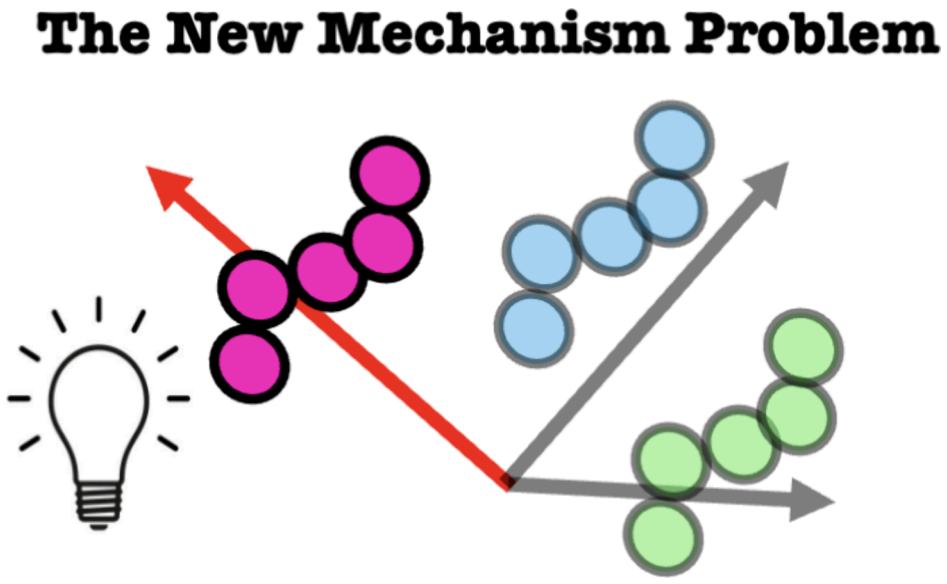


Figure 6. The new mechanism problem.

high concentrations of catalyst are usually required. RAFT is highly solvent, temperature, and functional group tolerant, but possesses a comparatively small monomer scope and RAFT agents are typically colored and unstable. While in some specific chemical contexts an expert can likely navigate these heuristics to accomplish a practical retrosynthesis, the utility of tools capable of solving this challenge broadly across chemical and reaction space and/or for nonexperts would be helpful. Notably, there has been tremendous progress in the past decade in developing practical solutions to the organic synthetic retrosynthesis problem, which has a much larger set of mechanisms than polymer retrosynthesis.[195−199] Moreover, even if this challenge can be solved across an arbitrary subset of chemical space leading back to purchasable starting materials, to move out of our current quagmire of "forever polymers" we need to accomplish this retrosynthesis via green pathways to non-petroleum based and sustainable starting materials. Note that this problem is concerned with retrosynthesis using known polymerization pathways, in contrast to **The New Mechanism Problem** which is concerned with the identification of novel pathways.

Examples of Success:

- Demonstrate a computational workflow that suggests source molecules derived sustainably via biomass or metabolites.
- Provided a candidate monomer chemistry and desired macromolecular architecture (MW, dispersity, branching), pick the ideal polymerization strategy from existing options.
- Provided an input monomer chemistry, produce a probability of success associated with any one of a finite set of possible polymerization strategies.
- Establish a predictive capability for choosing a particular style of metal-catalyzed chain-growth polymerization (e.g., Suzuki-Miyaura,[200] Heck,[201] Stille[202]).
- Enumerate the space of monomers amenable to catalyst transfer polymerization.[203,204]

6. The New Polymerization Mechanism Problem.

Statement: *Provided an idealized set of qualities for a polymerization, identify new polymerization reactions that meet these requirements.*

Synopsis: The ideal polymerization process should exhibit the following qualities: broad monomer tolerance, color free, metal (or even catalyst) free, inexpensive, environmentally friendly, works in aqueous media near room temperature, end groups can enable further functionalization, high MW control, controllable dispersity, recyclable/depolymerizable. The achievement of such a polymerization procedure is a holy grail of polymer chemistry and the likelihood of one polymerization meeting all these criteria is low. With that said, it is unlikely that we have already discovered all possible polymerization mechanisms compatible with organic polymer chemistries. While solution of **The Retrosynthesis Problem** will allow for selection of optimal existing pathways for polymerization, new polymerization mechanisms are needed to expand the scope of possible polymers and their architectures and molecular weight distributions (Figure 6). Step-function changes in defining polymerization reactions occur periodically in the literature; recent interest in dynamic and/or noncovalently bonded polymers[205] appears the most recent manifestation of relevance to polymerization reactions, but the future is rife with possibilities. One potential pathway forward is to mine existing databases of organic reactions[206] for compatibility with the requirements of polymerization. There is also the potential to revisit canonical organic chemistry reactions that may be underutilized relative to many of the more popular mechanisms (e.g., see advances in Diels−Alder[207,208] and other click chemistries for polymerization[209]). Of particular interest toward advancing polymer science is the ability to do new polymerization chemistry safely and not under extreme or nonideal settings (see, for example, the recent emergence of frontal polymerization techniques[210]).

Examples of Success:

- Computationally identify a new reaction mechanism, not present in the organic chemistry literature, conducive to polymerization.
- Identify a critical weakness of an existing reaction mechanism, and leverage computation to overcome this weakness through molecular design.

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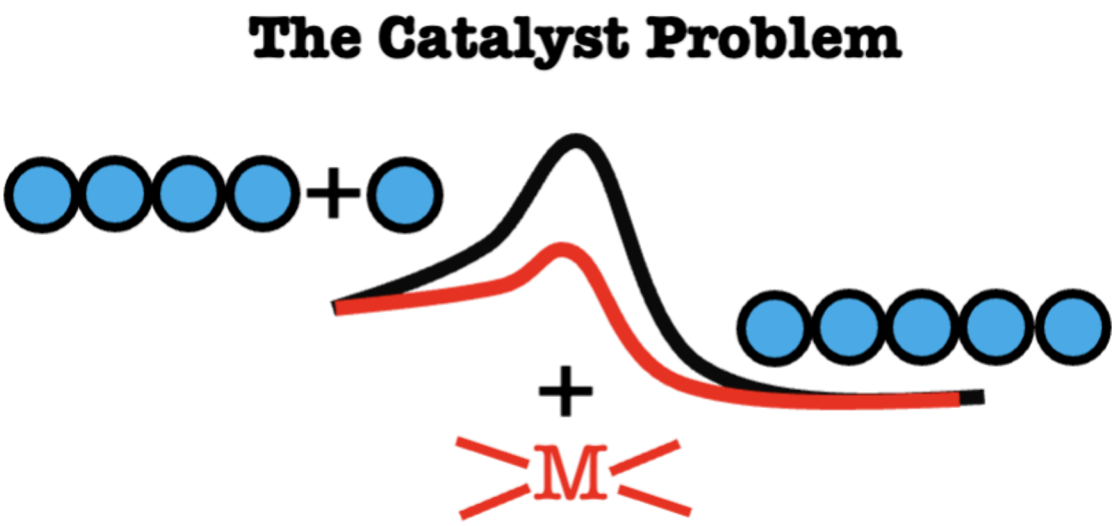


Figure 7. The catalyst problem.

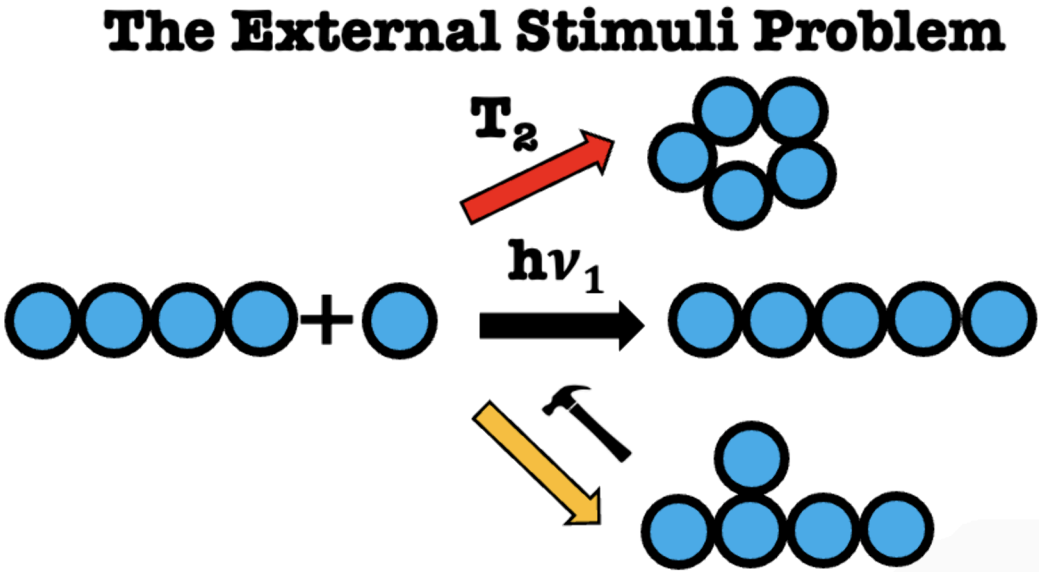


Figure 8. The external stimuli problem.

7. The Catalyst Problem. Statement: *Provided a monomer chemistry and polymerization mechanism, inversely optimize a catalyst.*

Synopsis: Although workable heuristics exist for many classes of polymerization, there is potentially large scope for optimizing catalysts with respect to specific monomer chemistries and polymer distribution targets (Figure 7). The explicit retrosynthetic suggestion of polymerization catalysts is not presently available in the community, and could have tremendous impact for nonexperts. Such an effort seems feasible due to the reduced scope of reactions and substrates for polymer chemistry relative to general organic synthetic chemistry, which may imply a high probability of success for already mature computational catalyst design strategies. Pressingly, there is the need to develop computational aids for the prediction of stereoselective polymerization catalysts, which have occupied tremendous experimental focus over the decades.[211,212] Particular focus moving forward could continue the strong recent momentum in metal free catalyst design for ROMP, ATRP, and RAFT chemistries. Moreover, there is the tantalizing potential to leverage computation to design new polymer chemistries that can proceed uncatalyzed, for which only a few cases exist (e.g., Nylon-6,6). In the longer-term, one can envision designing dynamic and stimuli-responsive catalysts[213] that can imbue polymers with new functionality. At the frontier of theory, this will require chemical physicists integrating systems theory, nonequilibrium statistical mechanics, and bioinspired computation to achieve advances. **The Catalyst Problem** is effectively a subproblem of the more broadly themed **The External Stimuli Problem**. However, the ubiquity and practicality of polymerization catalysts warrants its independence. The quantitative success of QC in modeling catalyzed organometallic reaction sequences suggests that the electronic structure and conformational sampling challenges are no longer prohibitive for this problem. However, modeling specific catalysts is fundamentally different from inversely optimizing over catalyst space. Near-term solutions might couple catalyst-specific characterizations with genetic algorithms to search over catalyst space. Longer-term, coupling generative models for sampling catalysts, active learning for prioritizing characterizations, and intelligently sharing catalyst information across optimization campaigns are

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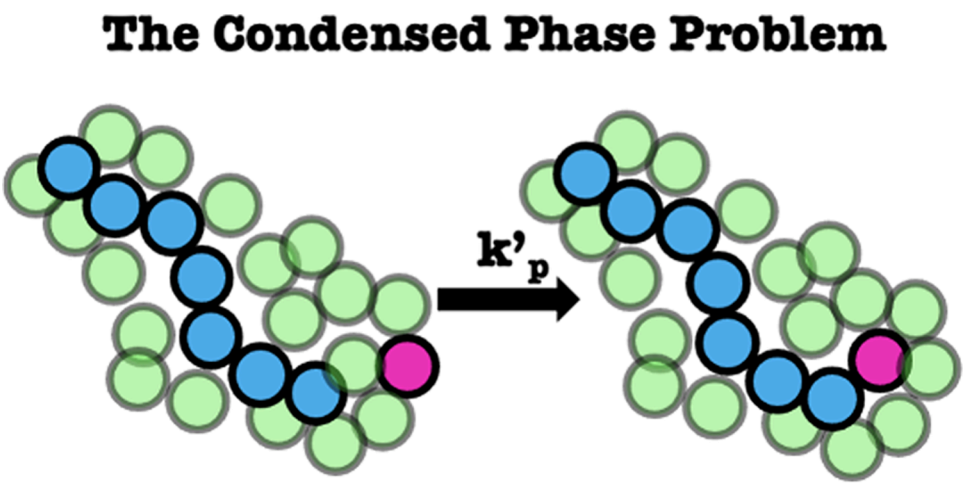


Figure 9. The condensed phase problem.

all active areas of research that might yield a step change in addressing this problem.

Examples of Success:

- Optimize a catalyst scaffold to maximize polymerization yield.
- Optimize a catalyst scaffold to maximize stereoselectivity and/or regioselectivity in a polymerization.
- Computationally engineer a new class of organic catalysts for anionic polymerizations.
- Engineer a molecular catalyst that can be turned on or off by an external stimuli with high specificity.

8. The External Stimuli Problem. Statement: *Provided an external stimulus (e.g., light, heat, mechanical stress, radiation) applied to a polymerization reaction, quantitatively predict its effect.*

Synopsis: The use of external stimuli to modulate polymerization reactions has been a popular strategy in polymer chemistry in the past decade (Figure 8).[163] While an immediately impactful (and nearly absent from the literature) computational effort would involve the use of excited-state QC[214,215] to predict light-induced modulations in reactivity, there is the potential for longer term and broader scope developments in chemical theory to dramatically impact externally controlled polymerization approaches. One potentially interesting target concerns how to model property changes in a material after synthesis upon exposure to external stimuli. The most well-trodden areas here computationally are likely the modeling of organic materials with embedded photoswitchable molecules such as azobenzene,[216,217] and the modeling of polymers that take advantage of mechanochemistry.[218] However, the ability to model light, heat, radiation, or mechanically induced changes in local bonding patterns could have a dramatic impact in the community. More broadly, while externally modulated polymerizations have been demonstrated in many contexts, there are almost no manifestations that show fast and specific actuations of synthetic polymers capable of being modified by an external stimulus (analogous to biopolymer synthesis in nature). What is almost certainly needed to achieve this paradigm shift can likely not be achieved by a single polymer entity, and must be induced via a systems level approach that leverages chemical fuel to drive a reaction out of equilibrium. Solutions to this problem will require multiscale strategies[219−222] that couple electronic structure characterizations with polymer mechanics occurring over potentially much longer time scales.[223] Recent methods like electronic coarse-graining[224−226] might yield a solution to this problem with a unified model. Alternatively, coupled approaches that use the results from atomistic modeling as inputs to classical or continuum level simulations are already relatively mature and could be adapted to this problem. At the computational level, this challenge rapidly falls into the scope of nonequilibrium statistical mechanical methodologies that have emerged in many driven and switchable systems.

Examples of Success:

- Use excited state QC methods to model the excited state reactivity of chain growth polymers.
- Develop a nonequilibrium polymerization strategy that uses chemical fuel.
- Predict how a polymerization reaction can be modulated via external mechanical force.[227]

9. The Condensed Phase Problem. Statement: *Provided a set of solvents, counterions, and/or specific molecular environment, predict the effects on the polymerization outcome.*

Synopsis: Condensed phase effects represent a grand challenge across theoretical science, and their manifestation in the modeling of polymer reactivity and synthesis is no exception (Figure 9). Condensed phase effects are strongest for ionic polymerizations, being a notable limiting factor on the scope of possible monomers for cationic polymerizations. Ionic polymerizations also present significant constraints on solvent selection (e.g., for cationic polymerizations one must avoid nucleophilic solvents). While continuum solvent models do a reasonable job in describing radical polymerizations in nonpolar or slightly polar solvents, their limitations for strongly ionic systems and protic solvents are well-documented.[228] The solution to the deficiencies of continuum models is an all-atom QM treatment of solvation, the cost of which limits such approaches to post facto assessments of existing experimental systems, and forbids their use in computational discovery campaigns. In the context of polymer synthesis, condensed phase effects also critically manifest in solubility, as high MW polymerizations can be solubility limited (e.g., the early precipitation of polyurethane in xylene and chlorobenzene, but not in DMSO). COSMO-RS is a gold standard for computational solubility prediction in molecules, but still lacks sufficient accuracy for application to polymers, due to the length dependent propagation of energetic errors into the solubility estimate.[229,230] To credibly address this

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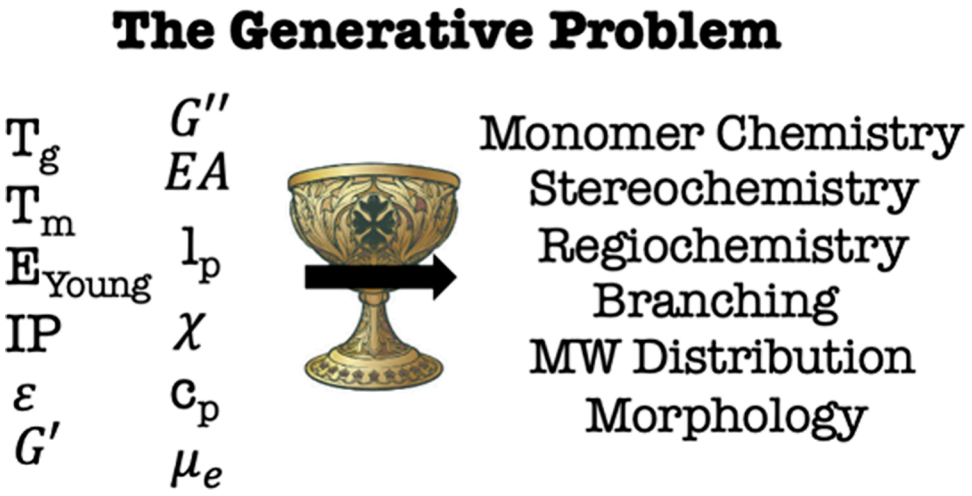


Figure 10. The generative problem.

will ultimately require an explicit treatment of many degrees of freedom over large length scales. Although this was once an impossible combination, the recent emergence of MLIPs with DFT-level accuracy, but classical force-field costs, now make this possible. While the majority of this perspective operates under the guise of predicting intrinsic reaction rates, diffusion-limited reactivity is a known challenge,[141,231,232] especially at high molecular weight, not to mention the challenges at larger length scales of reactor heterogeneity that are excluded from our molecularly focused discussion.

Examples of Success:

- Establish a simulation protocol capable of estimating the nature of the ion pair state involved in propagation, and its associated free-energy profile.
- Establish a simulation protocol capable of estimating protic solvent effects on radical polymerization that is scalable to computational discovery campaigns.
- Estimate polymer solubility from a monomer chemistry and polymer topology to predict solubility-limited polymerization.[233]
- Predict how polymer reactivity is affected in a solid-phase synthetic environment (e.g., on a resin) relative to the solution-phase.
- Predict cross-linking in a ternary polymer blend with full incorporation of structural dynamics.

10. The Generative Problem. Statement: *Provided a targeted set of polymer properties, generate polymer chemistries accounting for intrinsic heterogeneities of polymeric materials.*

Synopsis: The holy grail of computational polymer design. Even if any chemical repeat unit could be synthesized, polymer properties are dictated by more than their repeat unit(s). Molecular weight, dispersity, defects, stereochemistry, branching and processing protocols all contribute strongly to a polymer's ultimate behavior. At the very least, practical generative models for polymers should be equipped to handle arbitrary chemistry, macromolecular architecture, and intrinsic stochasticity (Figure 10). Some works[234−239] have begun to broach this topic, but a method that can incorporate both chemistry and polymer architecture in a fully generative architecture is still lacking. It is likely that no such generative architectures will be achievable without either a thoughtful combination of physics-based priors and/or a vast amount of (open) polymer structure−property data far beyond anything that exists today. Moreover, the training data for such a generative model will likely be derived from the methods that are developed to address the foregoing problems highlighted in this perspective. However, on the computational front, the need for pursuing such generative models is clear, and constitutes an important future direction for computational polymer design, especially when integrating considerations of synthetic viability.

Examples of Success:

- Develop a generative ML architecture that can simultaneously incorporate at least two of the following four attributes: chemical repeat unit structure, structural heterogeneity (e.g., branching, dispersity), conformational heterogeneity, synthetic accessibility.
- Develop a generative ML architecture that can enforce physics-based priors derived from existing knowledge in polymer theory, and can be complemented via a chemically specific, data-driven approach.

THE ROLE OF DATA INFRASTRUCTURE

While the majority of the discussion has been motivated by the inherent difficult of experimentally measuring polymer reactivities in comparison to computational approaches, this is not intended to downplay the importance of experimental benchmarks for calibrating reactivity prediction methods. The importance of uncertainty quantified experimental databases of reactivities for the calibration of computational campaigns cannot be overstated. In the context of reactivity ratios in copolymerization, there is a surprisingly large amount of available literature data with different copolymerization chemistries, but limited quantification of the uncertainty or noise across these measurements. While one can speculate on such uncertainties by consideration of best practices,[240−242] the lack of high quality data sets for polymer reactivity is a limiting point for any computational approach. Similar problems reoccur across every avenue of polymer reactivity studies, influencing every problem described in this perspective.

Moving forward it is important to establish and incentivize improved data infrastructures, especially for experimental work, to support the understanding and modeling polymer reactivity. For example, developing polymer specific reactivity databases via inclusion in platforms such as ORD,[206] CRIPT,[243] Reaxys, and the IUPAC subcommittee curated database of radical

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polymerization kinetics,[181] could significantly advance data accessibility and integration within the field.

CONCLUSIONS

Polymer chemistry stands to benefit from recent developments in computational reactivity methodologies, as well as the broadening of what constitutes "polymer theory" in the polymer science community. We have outlined ten challenge problems for polymer reactivity prediction, though we have every expectation that more will emerge in the coming years. The solution of these problems will enable a future of rational synthetic polymer design in which synthetic accessibility and end-of-life degradation can be naturally integrated into the design process. Provided the enormous number of researchers presently focusing on the computational design of classical polymer properties at large length scales, it is perhaps appropriate to adapt Feynman's quotation: "There is Plenty of Room at the Bottom" for the computational design of polymers.

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Notes

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